FIX_EVIDENCE

Translate-arg DATA-LOSS bug class — FIX + RED→GREEN evidence (§11.4.115/§11.4.135/§11.4.146)

Revision: 1 **Last modified:** 2026-06-16T00:00:00Z **Fix of:** the 6 confirmed-broken `.Translate(ctx,text,prompt)` call sites in `docs/qa/translate_arg_audit_20260616-153353/AUDIT.md` + `docs/qa/translate_arg_audit_completion_20260616-154307/COMPLETION.md` + `docs/qa/bug_markdown_empty_payload_rootcause_20260616-152640/FINDING.md`. **Method:** superpowers:systematic-debugging Phase 4 + §11.4.6 FACT-only + §11.4.69 + §11.4.115.

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1. Fix design (two choke-points cover all 6 sites)

Seam A — `pkg/translator/llm/openai.go` `OpenAIClient.Translate` (defense-in-depth)

Covers the 4 EMPTY-PAYLOAD sites (#1 `cmd/markdown-translator/main.go:244`, #2/#3/#4 `pkg/preparation/coordinator.go:290/361/424`) that reach a RAW OpenAI-compatible client with an empty 2nd arg.

- when `prompt` is empty/whitespace → fall back to the content-bearing 1st arg `text`;
- when BOTH are empty/whitespace → return an explicit error (§11.4.69), never send an empty user message that the model answers with boilerplate stored as a translation.
- Embedded by DeepSeek/Groq/etc., so the guard covers every OpenAI-compatible provider.

Seam B — `pkg/bridge/bridge.go` `clientTranslator.Translate` (compose choke-point)

Covers the 3 WRONG-CONTENT sites (#5 `pkg/verification/polisher.go:563`, #8/#9 `pkg/coordination/multi_llm.go:445/529`) that reach the verbatim-delegating `clientTranslator` via `Translate(ctx, REAL_CONTENT, LABEL)`.

- the content-bearing 1st arg `text` is always placed in the body the raw client actually sends (its 2nd arg), mirroring the `bridge.go:369` `Translate(ctx,"",full)` convention; a non-empty `contextStr` is appended as a labelled `Context:` hint, never substituted for the content;
- empty `contextStr` → the content is sent verbatim (preserves the preparation-ensemble `(prompt, "")` shape #2/#3/#4 reaching `clientTranslator` — no double-wrapping);
- empty/whitespace content → explicit error (§11.4.69).

Seam C — `cmd/markdown-translator/main.go:244` (belt-and-suspenders caller)

Build a real translation instruction embedding the block text and pass it as the 2nd arg (mirrors `pkg/markdown/simple_workflow.go`), so the cmd no longer relies solely on the Seam-A fallback.

§11.4.6 correction to the dispatch's "fix the 4 empty-payload callers to pass

content in arg-2 too": the preparation callers #2/#3/#4 pass a COMPLETE analysis prompt in arg-1 with an empty arg-2. Rewriting them to `("", prompt)` would (a) make `clientTranslator` refuse empty content and (b) make `*LLMTranslator` return early on empty `text` — BREAKING both paths. They are FIXED by the two choke-points and correctly LEFT as `(prompt, "")`. Likewise #5 polisher is FIXED by Seam B (clientTranslator now composes the verification `prompt` + `location`), so polisher.go:563 is unchanged — editing it to `("", …)` would break the `*LLMTranslator` default path.

Which of the 6 sites is fixed by what

#	site	class	fixed by
1	markdown-translator/ main.go:244	empty-payload	Seam A fallback + Seam C real prompt
2	preparation/ coordinator.go:290	empty-payload	Seam A fallback (raw) / Seam B verbatim-prompt (ensemble)
3	preparation/ coordinator.go:361	empty-payload	same as #2
4	preparation/ coordinator.go:424	empty-payload	same as #2
5	verification/ polisher.go:563	wrong-content	Seam B compose (ensemble path)

#	site	class	fixed by
8	coordination/ multi_llm.go:445	wrong-content	Seam B compose
9	coordination/ multi_llm.go:529	wrong-content	Seam B compose

(#11 server.go:196 DEAD-CODE, #22 notes.go:166 SAFE — per COMPLETION.md, untouched.)

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2. RED→GREEN evidence (captured this session, httptest, no API key)

Seam A — `TestOpenAITranslate_EmptyPayloadGuard` (+ `TestOpenAITranslate_RefuseAllEmpty`)

```
GREEN (fixed, default):      ok   pkg/translator/llm
RED_MODE=1 on PRE-FIX openai.go (stashed fix): ok  -> defect
reproduced (empty msg sent, boilerplate returned)
RED_MODE=1 on FIXED openai.go:                 FAIL -> content
"The old man walked along the shore at dawn." now reaches model
```

Seam B — `TestClientTranslator_ComposesContentNotLabel` (+ EmptyContext + RefusesEmpty)

```
GREEN (fixed, default):      ok   pkg/bridge
RED_MODE=1 on PRE-FIX bridge.go (stashed fix): ok  -> defect
reproduced (label "Section content" sent, real content dropped)
RED_MODE=1 on FIXED bridge.go:                 FAIL -> real
content REAL_SECTION_CONTENT_THAT_MUST_BE_TRANSLATED now reaches
model
```

The RED tests are registered as permanent §11.4.135 standing guards (default `RED_MODE=0`).

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3. Validation

```
go build ./...                                    -> exit 0
go test ./pkg/translator/llm/ ./pkg/bridge/ \
        ./pkg/coordination/ ./pkg/verification/ \
        ./pkg/preparation/ ./cmd/markdown-translator/ \
        ./pkg/markdown/ -count=1                   -> all ok
```

No SAFE site broke: `TestOpenAITranslate_RequestShapeAndAuth` (asserts the non-empty 2nd-arg user message reaches the wire verbatim) still passes — the Seam-A fallback is inert for non-empty 2nd args, and the Seam-B compose only fires inside `clientTranslator`.

4. §11.4.6 honesty boundary

- PROVEN FACT (wire repro, both polarities, both seams): empty-payload fixed at the raw client; wrong-content fixed at the clientTranslator choke-point; no SAFE site regressed.
- The end-to-end #8/#9 production seam is the real `bridge.clientTranslator` (the exact type `instance.Translator` holds in the factory ensemble path per COMPLETION.md §1) — the Seam-B test drives that real type, not a stub.